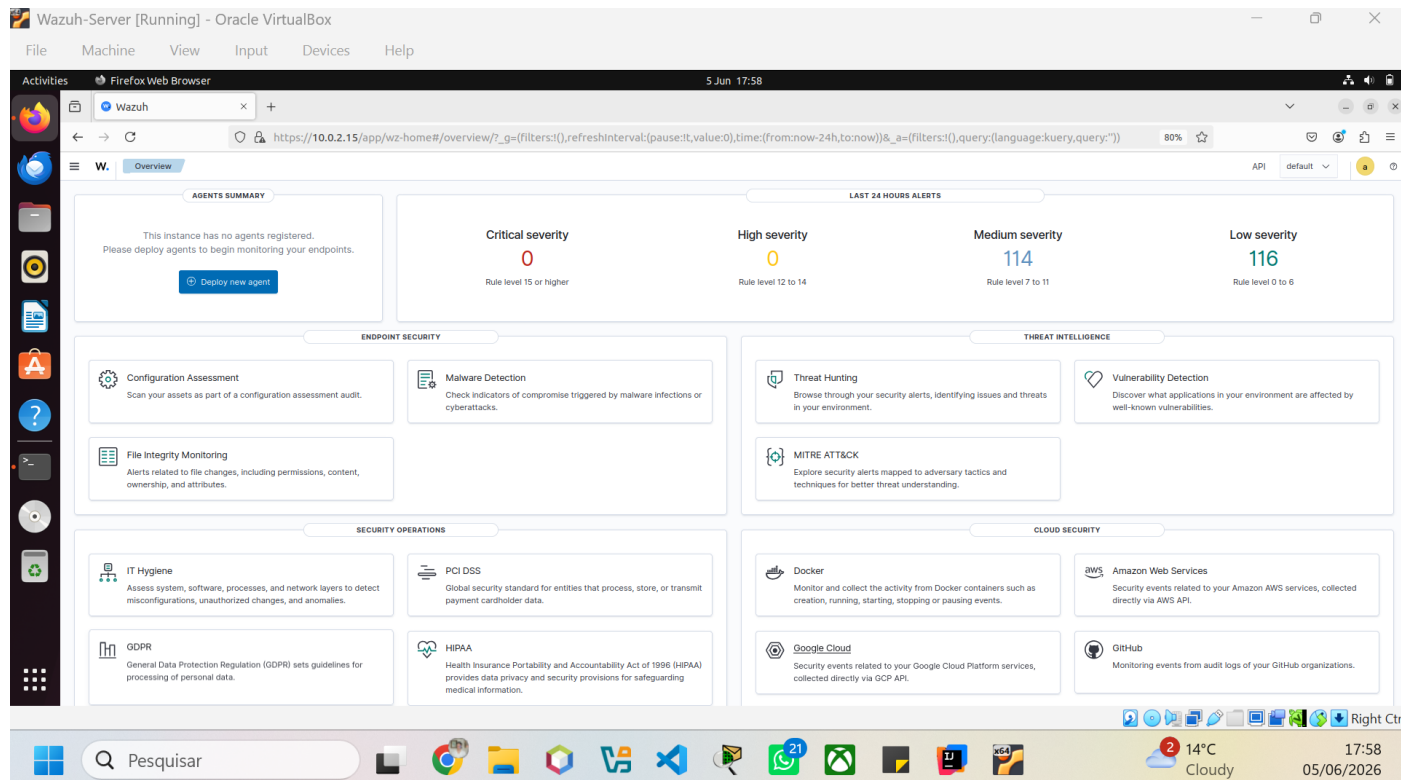


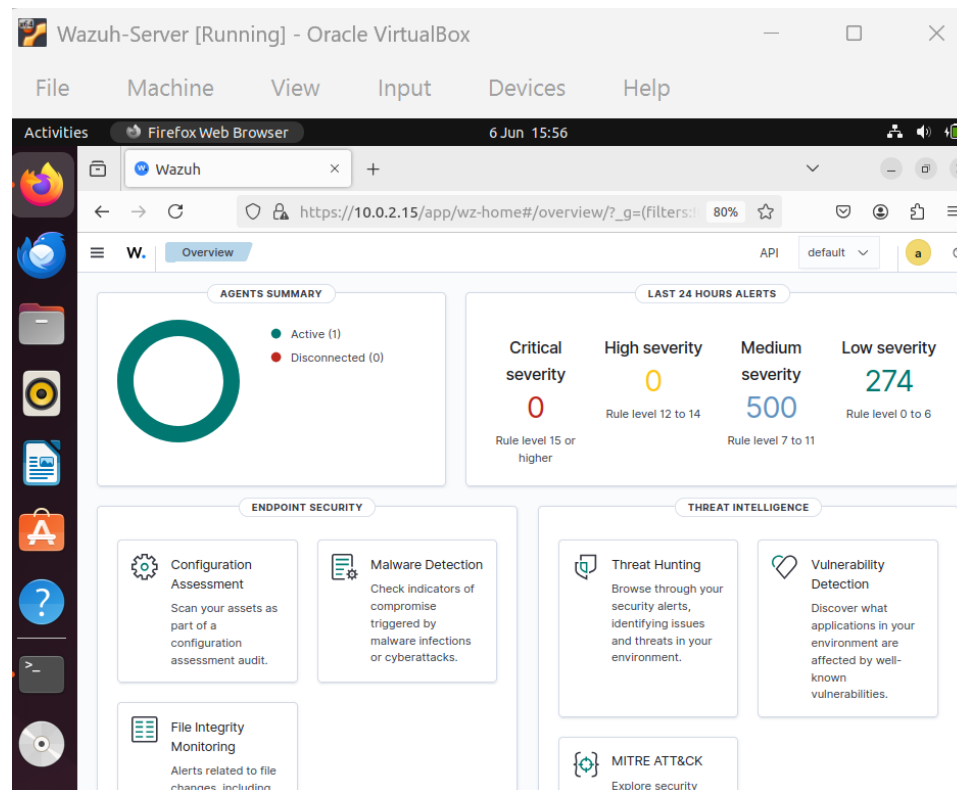
Project Introduction

Business Goal: Build a functional SIEM lab using Wazuh to simulate a SOC environment. The objective was to centralize logs, onboard endpoints, perform threat hunting, detect suspicious activity, and other functionalities.



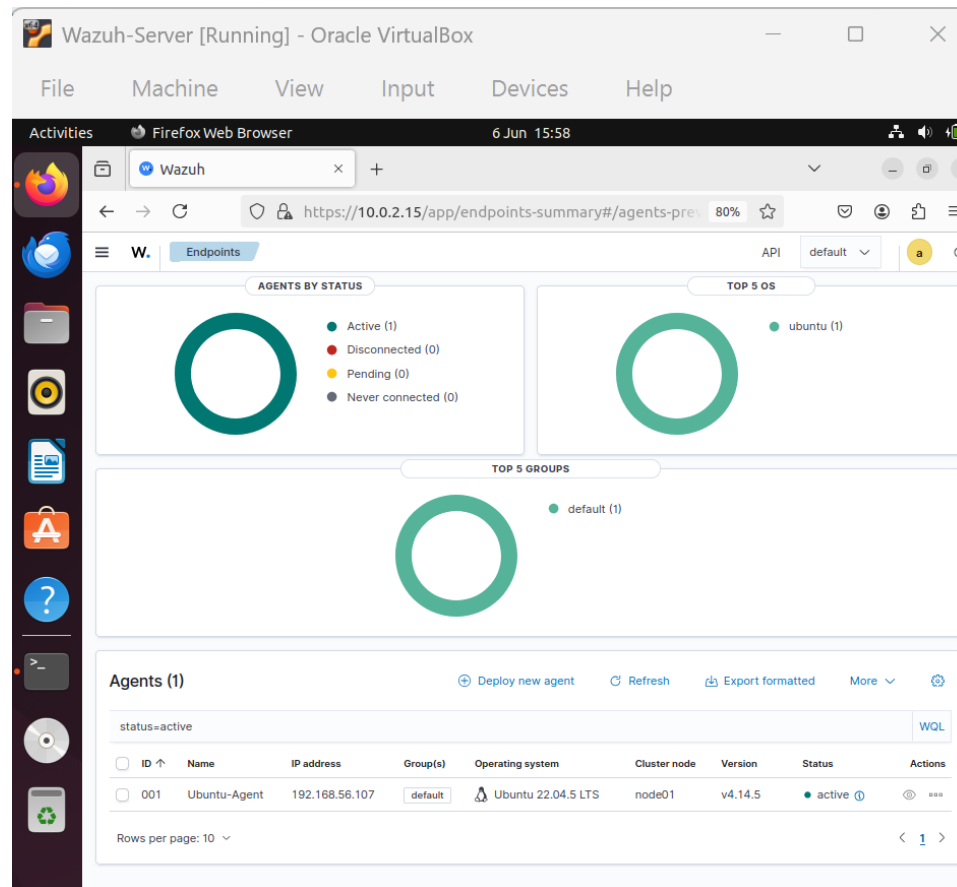
Wazuh Deployment Successful

After deploying Wazuh 4.14.5 on Ubuntu 22.04, the dashboard became operational. This confirmed that the Manager, Indexer, and Dashboard components were functioning correctly. The environment was now ready to receive endpoint telemetry.



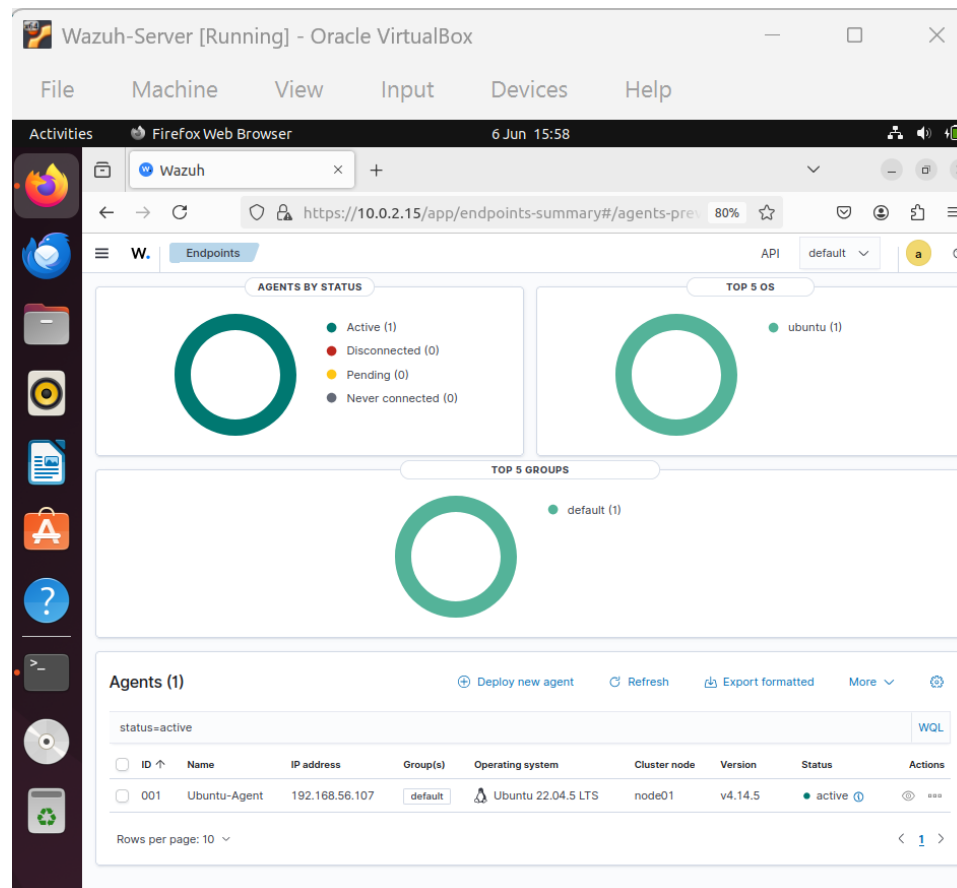
Endpoint Enrollment

A second Ubuntu VM was configured as a monitored endpoint. Multiple networking and enrollment issues were investigated and resolved. The endpoint was successfully registered with the Wazuh Manager.



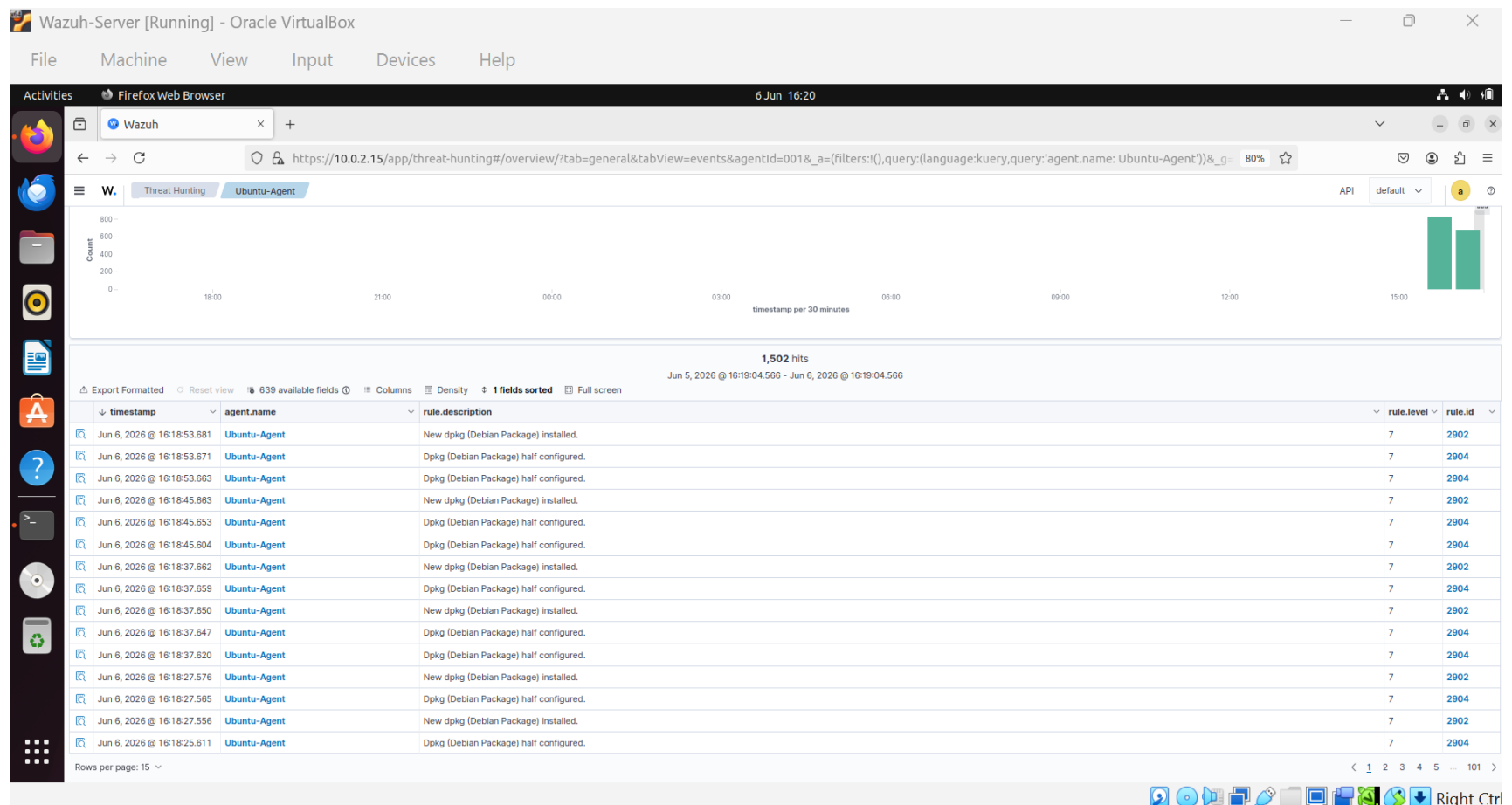
Endpoint Visibility Confirmed

The Endpoints dashboard confirms successful onboarding. Agent ID 001 is active and communicating with the manager. This validates connectivity, agent registration, and telemetry collection.



Threat Hunting Validation

Threat Hunting was used to validate log ingestion. Package installation and system activity logs generated by the endpoint were indexed and searchable. This proves successful event collection and storage.



User Creation Detection

A new local user (hackerA) was intentionally created to generate a security-relevant event. Wazuh immediately detected the action and recorded username, UID, GID, shell, and home directory details.

The screenshot displays the Wazuh Server interface within an Oracle VirtualBox window. The main dashboard shows a list of events for the 'hackerA' agent. A 'Document Details' panel on the right provides a structured view of the event data.

Wazuh-Server [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Firefox Web Browser 6 Jun 16:26

Wazuh Threat Hunting Ubuntu-Agent

Dashboard Events

hackerA

manager.name: Wazuh-Server agent.id: 001 Add filter

Count

timestamp

Jun 5, 2026 @ 16:25:39 247

Export Formatted Reset view 639 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description
Jun 6, 2026 @ 16:22:50.340	Ubuntu-Agent	New user added to the system.
Jun 6, 2026 @ 16:22:50.338	Ubuntu-Agent	New group added to the system.
Jun 6, 2026 @ 16:22:50.332	Ubuntu-Agent	Successful sudo to ROOT executed.

Document Details

View surrounding documents View single document

Table	JSON
._index	wazuh-alerts-4.x-2026.06.06
agent.id	001
agent.ip	192.168.56.107
agent.name	Ubuntu-Agent
data.dstuser	hackerA
data.gid	1002
data.home	/home/hackerA
data.shell	/bin/sh,
data.uid	1002
decoder.name	useradd
decoder.parent	useradd
full_log	Jun 06 15:22:48 Ubuntu-Agent useradd[98873]: new user: name=hackerA, UID=1002, GID=1002, home=/home/hackerA, shell=/bin/sh, from=/dev/pts/0
id	1788759370.1555050
input.type	log
location	journalid
manager.name	Wazuh-Server
predecoder.hostname	Ubuntu-Agent
predecoder.program_name	useradd
predecoder.timestamp	Jun 06 15:22:48
rule.description	New user added to the system.
rule.firedtimes	1
rule.gdpr	IV_35.7.d, IV_32.2
rule.gpg13	4.13

Right Ctrl

Authentication Failure Detection

Failed authentication attempts were generated using incorrect sudo credentials. Wazuh identified the event as a PAM authentication failure and captured user, session, source host, and timestamp information.

The screenshot displays the Wazuh Server interface within an Oracle VirtualBox environment. The browser window shows the Wazuh dashboard with the 'Threat Hunting' tab selected. The 'Events' section is active, displaying a list of events. A specific event is highlighted, showing details for a PAM authentication failure. The event details are as follows:

Field	Value
_index	wazuh-alerts-4.x-2020.06.06
agent.id	001
agent.ip	192.168.56.107
agent.name	Ubuntu-Agent
data.destuser	agent
data.euid	0
data.srcuser	agent
data.tty	/dev/pts/1
data.uid	1000
decoder.name	pam
full_log	Jun 06 15:23:27 Ubuntu-Agent sudo[99939]: pam_unix(sudo:auth): authentication failure; logname= uid=1000 euid=0 tty=/dev/pts/1 ruser=agent rhost= user=agent
id	1788759488.1570586
input.type	log
location	journalid
manager.name	Wazuh-Server
predecoder.hostname	Ubuntu-Agent
predecoder.program_name	sudo
predecoder.timestamp	Jun 06 15:23:27
rule.description	PAM: User login failed.
rule.firedtimes	1
rule.gdpr	IV.35.7.d, IV.32.2
rule.gpg13	7.8
rule.groups	pam, syslog, authentication_failed

Security Configuration Assessment

The CIS Ubuntu benchmark checks demonstrate Wazuh's ability to perform security posture assessments. This helps identify configuration weaknesses, hardening opportunities, and compliance gaps.

Wazuh-Server [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Firefox Web Browser 6 Jun 16:27

Wazuh Threat Hunting Ubuntu-Agent

Dashboard Events

failed

manager.name: Wazuh-Server agent.id: 001

Count

timestamp

Jun 6, 2026 @ 15:47:01.830 Ubuntu-Agent CIS Ubuntu Linux 22.04 LTS Benchmark v2.0.0: Ensure permissions on /etc/security/opaswds

Jun 6, 2026 @ 15:47:01.923 Ubuntu-Agent CIS Ubuntu Linux 22.04 LTS Benchmark v2.0.0: Ensure permissions on /etc/shells are co

Jun 6, 2026 @ 15:47:01.904 Ubuntu-Agent CIS Ubuntu Linux 22.04 LTS Benchmark v2.0.0: Ensure system warns when audit logs ar

Jun 6, 2026 @ 15:47:01.885 Ubuntu-Agent CIS Ubuntu Linux 22.04 LTS Benchmark v2.0.0: Ensure systemd-journal-upload is enable

Jun 6, 2026 @ 15:47:01.861 Ubuntu-Agent CIS Ubuntu Linux 22.04 LTS Benchmark v2.0.0: Ensure filesystem integrity is regularly c

Jun 6, 2026 @ 15:47:01.809 Ubuntu-Agent CIS Ubuntu Linux 22.04 LTS Benchmark v2.0.0: Ensure root password is set.

Jun 6, 2026 @ 15:47:01.733 Ubuntu-Agent CIS Ubuntu Linux 22.04 LTS Benchmark v2.0.0: Ensure pam_unix does not include nullok

Jun 6, 2026 @ 15:47:01.731 Ubuntu-Agent CIS Ubuntu Linux 22.04 LTS Benchmark v2.0.0: Ensure pam_pwhistory includes use_aut

Jun 6, 2026 @ 15:47:01.726 Ubuntu-Agent CIS Ubuntu Linux 22.04 LTS Benchmark v2.0.0: Ensure password history is enforced for

Jun 6, 2026 @ 15:47:01.711 Ubuntu-Agent CIS Ubuntu Linux 22.04 LTS Benchmark v2.0.0: Ensure password history remember is co

Jun 6, 2026 @ 15:47:01.693 Ubuntu-Agent CIS Ubuntu Linux 22.04 LTS Benchmark v2.0.0: Ensure password quality checking is en

Document Details

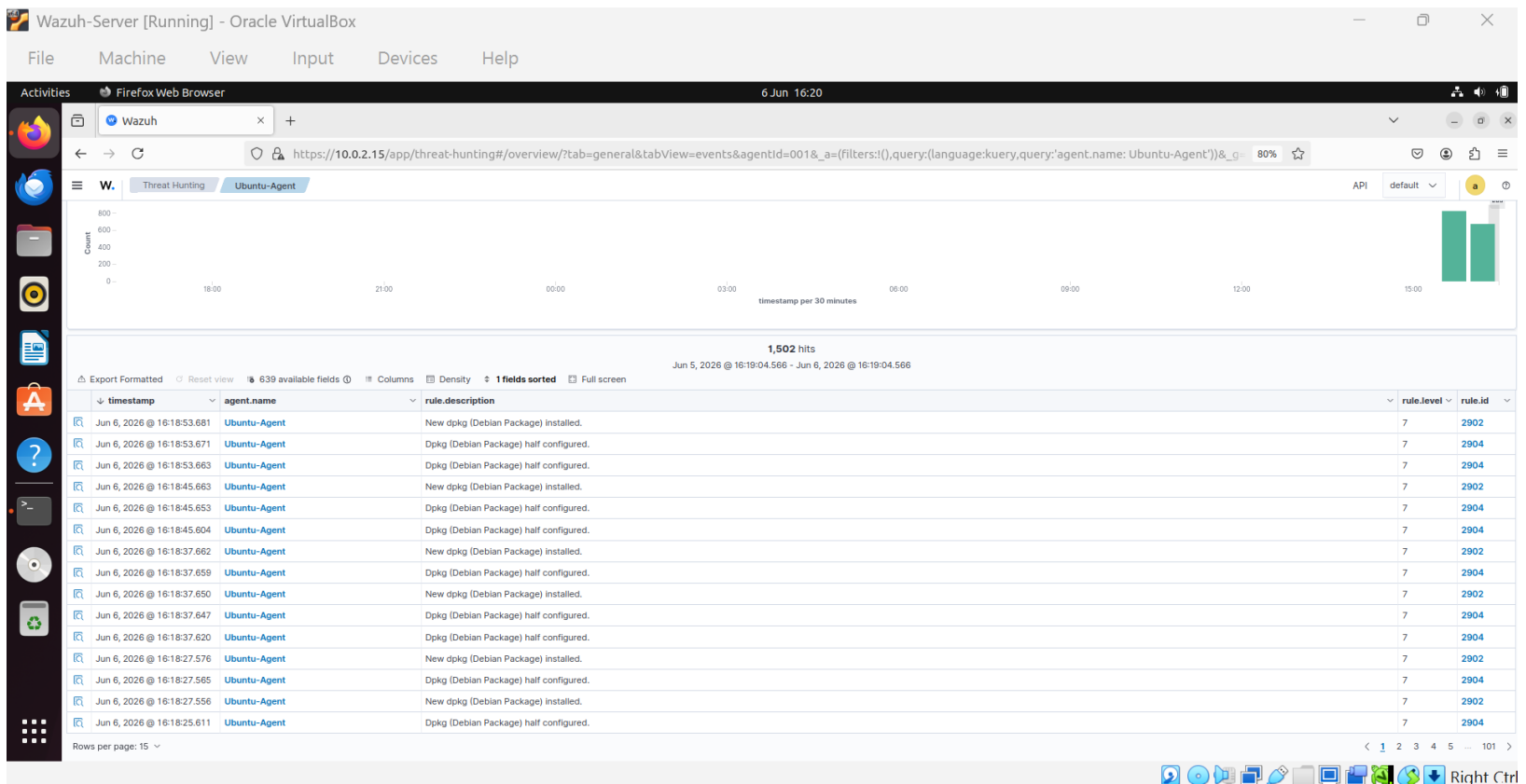
View surrounding documents View single document

Table JSON

```
{
  "_index": "wazuh-alerts-4.x-2026.06.06",
  "agent.id": "001",
  "agent.ip": "192.168.56.107",
  "agent.name": "Ubuntu-Agent",
  "data.sca.check.command": "stat -Lc \"\n %a %u %g %0\" /etc/security/opasswds",
  "data.sca.check.compliance.ci": "7.1.10",
  "data.sca.check.compliance.ci.s_csc_v7": "14.6",
  "data.sca.check.compliance.ci.s_csc_v8": "3.3",
  "data.sca.check.compliance.cm": "AC.L1-3.1.1,AC.L1-3.1.2,AC.L2-3.1.3,AC.L2-3.1.5,MP.L2-3.8.2",
  "data.sca.check.compliance.hi": "164.388(a)(3)(i),164.388(a)(3)(is)(A),164.312(a)(1)",
  "data.sca.check.compliance.is": "A.9.1.1",
  "data.sca.check.compliance.mi": "M1822",
  "data.sca.check.compliance.mi.tre_mitigations": "TA0005",
  "data.sca.check.compliance.mi.tre_tactics": "T1083,T1083.008,T1222,T1222.002",
  "data.sca.check.compliance.ni": "AC-5,AC-6",
  "data.sca.check.compliance.pc": "7.1.7.1.1,7.1.2.7.1.3",
  "data.sca.check.compliance.pc.i_dss_v3.2.1": "1.3.1.7.1",
  "data.sca.check.compliance.pc.i_dss_v4.0": "1.3.1.7.1"
}
```

SOC Analyst Activities Demonstrated

This project demonstrates endpoint onboarding, alert monitoring, threat hunting, log analysis, authentication monitoring, account monitoring, Linux administration, SIEM operations, and incident investigation workflows.



Project Outcome

The final environment successfully collected and analyzed endpoint events. The SIEM detected user creation activity, authentication failures, package installations, and configuration assessment findings. The project provided practical SOC-focused experience using industry-relevant tooling.

